

Ying Xiang Seah

Mixed Reality Developer

Computer Science and Design Undergraduate @ Singapore University of Technology and Design



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Work Experience

Mixed Reality Developer • HelloHolo • Singapore

Full-time • Feb 2024 - Aug 2024 • 6 mo

- Spearheaded development of an internal UI Toolkit in Unity to streamline XR workflows, replacing the old and unsupported Mixed Reality Toolkit (MRTK).
 - Advanced **Unity editor scripting**.
 - Created comprehensive API documentation using DocFX.
- Served as **trainer** for DSTA BrainHack XRperience 2024 for over 120 participants
 - Created and delivering **Unity** and **C#** instructional materials for **cross-platform XR** content creation.
 - Demonstrated Apple Vision Pro's capabilities and provided development support.

Part-time • Oct 2020 - Oct 2022 • 2 yr

- Assistant trainer** for immersive media courses with over 150 students for several major local education institutions.
- Built a **digital twinning** experience for National University of Singapore's 5G Lab exhibition.
- Worked with Microsoft Egypt to deliver an interactive data visualization app that displays **dynamic real-time IOT data** on construction analytics for a prominent national project.
- Delivered multiple MR-based **training simulators** on HoloLens 2 for various prominent global corporations for their internal employee training initiatives.
- Assisted as a **technical consultant** for a student research project from a renowned local technological university, exploring the use of **digital twinning in MR** to aid engineers in testing and evaluating mechanical designs and assemblies from CAD software.

Projects

MUVE Virtual Museum Tour - Extended Reality Demo

- Coordinated team of 6 to deliver a **Virtual Museum** Game Demo in VR using Unity.
- Implemented custom **UI Toolkit editor tooling** to simplify visual feedback of 3D UI.
- Created functional virtual museum tour demo with **high quality assets** from public domain 3D scanned artefacts.
- Implemented voice narration guide using AI.
- Implemented fun minigame to reward knowledge of museum artefacts

FPGA Arcade Game and ALU design

- Designed a working **16-bit ALU** and **CPU** following MIT's Beta CPU ISA using Lucid HDL, deployed on the Xilinx Artix-7 **FPGA**.
- Wrote **graphics driver** for our custom CPU design to display graphics on an RGB LED matrix.
- Prototyped **game code in C** and manually transpiled C code to the Beta assembly language to run on our CPU implementation on the FPGA.

The Pyramid - Design Thinking Architecture

- Led team of 5 through design thinking phases to come up with a design project to improve an aspect of the local business park environment.
- Coordinated efforts to model and render our **3D architectural prototype** in Blender.
- Presented our functional and sustainability analysis on Figma.

Mycelium - Design Thinking Architecture

- Generative modelling and design of a cabin structure using **Rhino** and **Blender**.
- Environment design and rendering in **Unreal Engine 5**.

Skills

Full Professional Proficiency

- Java
- Python
- JavaScript
- HTML/CSS
- Unity
- C#
- C/C++
- Git
- TypeScript
- Figma

Decent Proficiency

- Lua
- Blender (3D modelling/rendering)

In Progress

- Unreal Engine 5
- Blender (texturing, rigging)

Soft Skills

- Presentations/Teaching
- Team leadership
- Adaptable
- Resourceful
- Proficient in Mandarin and English

Related Coursework

- Computational Structures
- Computer System Engineering
- Algorithms
- Information Systems
- Elements of Software Construction
- Machine Learning
- Applied Deep Learning
- Extended Reality
- Graphics and Visualisation
- Programming Language Concepts

Education

Bachelor of Engineering in Computer Science and Design

Singapore University of Technology and Design

2021-2025 • CGPA 4.50

Executive member of Cycling IG

- Organized and hosted weekly biking activities
- Managed club finances

Member of the SUTD XRCommunity

- Organized club meetings and presented topics on current events in XR (Extended Reality)

Hobbies & Interests

- Portrait and Travel Photography
- Road cycling
- VR gaming and game development
- Rollercoaster rides and whitewater rafting
- Avid manga reader